COMP1831 TECHNOLOGIES FOR ANTI-MONEY LAUNDERING AND FINANCIAL CRIME

Anti-Money Laundering As A Service Using Federate Learning

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Contents

1 Background Research 2
2. Overview of the Proposed System 2
3. Data Access Needs 4
4. Data Privacy and Governance 4
5. High Level Architecture 5
6. Tools and Methodologies 6
7. Conclusion 8
8. References 8

1 Background Research

The objective of Money Laundering (ML) is to create the impression that money from predicate criminal activities originates from legitimate sources, as articulated by (Demetis, 2018).It is further emphasised that the transformation of illicit funds into legal currency allows criminals to engage with the legitimate financial system to their advantage, thereby rendering them more freedom in the transfer funds to facilitate additional criminal activities. Despite difficulties in estimating the underground market size, ML is projected to represent between $1.5 trillion and $2.85 trillion each year, making it the third largest market globally. According to(Parveen, 2020), the ML activities in the United Kingdom is likely to exceed 15 billion pounds each year.

According to (Matanky-Becker and Cockbain, 2022), the Financial Action Task Force (FATF) support a definition of ML through a three-stage model comprising placement (the introduction of illicit funds into the financial system), layering (the creation of a separation between the illicit funds and the original crime), and integration (the illicit funds are reintroduced into the legal economy). However, he argues that this model is limited and outdated, thus posing potential risk, as practitioners who misinterpret ML maybe poorly prepared to prevent or investigate it. He claims the use of this model in research will be problematic due to its weal conceptual foundations and vague definitions.

A significant obstacle to enhancing Anti-Money Laundering (AML) protocols is the fragmentation of intelligence among financial institution. Factors like regulatory caution, competitive constraints and data privacy regulators (UK GDPR) hinder the exchange of sensitive financial and customer information between institutions. Accordingly, each institution operates mostly independently, using localised models based on limited and biased data pools. This isolated methodology reduces the overall efficiency of the financial systems in detecting complex laundering operations involving multiple parties or jurisdictions (HM Treasury, 2024).

In order to overcome this limitation, this report suggests the development of a conceptual AML. This methodology all financial institutions like banks Payment Service Providers (PSPs) and Fintech companies to work together to build a machine learning model that is aimed at detecting suspicious activity patterns without the need to exchange customer information and raw transaction data. By using federated learning, each organisation keeps ownership of its data while contribution to the improvement of a better model that draws from a wide range of contributors. This will offer an effective balance between collaborative intelligence and also adhering to data privacy regulators.

The proposed system will specifically assess risk involved in the layering phase of the ML phase when it involves a high likely hood of the illicit funds being conceded within intricate transaction networks. By promoting cross-institutional learning through federated methods, the proposed solution aims at identifying ML activity patterns that might not be discovered within isolated data pools

2. Overview of the Proposed System

The proposed system of an Anti-Money Laundering as a Service is aimed to bridge institutions in the identification of KL activities without the need of sharing sensitive customer or transaction information. The platform utilises a federated learning technique which is a machine learning technique allowing different institutions to cooperatively build a consensus model without sharing raw data. The institution which are involved will gain an improved detection due to their collective knowledge of the network while ensuring compliance with the strict privacy laws like the UK GDPR and Data Protection Act of 2018.

The system is aimed at regulated financial institutions that must put AML controls but encounter difficulties due to limited access to data and resources. This situation is particularly relevant to smaller PSPs and emerging fintech companies as they do not have adequate data pools to independently develop effective machine learning model. Furthermore, this system can provide collective intelligence to regulators like Financial Conduct Authority (FCA) and Payment System Regulator (PSA) for developing policies.

In terms of operation, the system will act as a distributed environment for model training. Each institution involved jeep its transaction and customer data in their own infrastructure and a local model aimed at identifying suspicious activities. After the model training, the institution will send encrypted model parameters (gradients or weights) to a trusted central aggregator. It is important that no unprocessed data is shared outside the institution. The central aggregator will use federated averaging to combine these parameters into a single global model, which is subsequently redistributed to all participating institutions. This iterative process enables the model to gain insights from a wider array of ML activity samples from across multiple institutions.

To guarantee both privacy and resilience, the system utilises a number of advanced technologies. Differential Privacy is used to update the model prior to their transmission, to ensure safeguarding against individual data entries to be unidentifiable within aggregated parameters (Dwork, 2006). At the aggregated step, Secure Multi Party Computation is implemented to restrict the central aggregator the access to individual model updates in their unencrypted form(Lindell, 2021). Additionally, an optional blockchain lay can be added to allow the creation of an immutable ledger of model updates and contributions to greatly improve both accountability and auditability.

The platform outputs differ depending on the stakeholder. For all participating financial institutions, the primary output is an improved AML detection model customised to fit their specific local context while benefiting from the shared intelligence. This model can easily be incorporated into their current transaction monitoring systems to produce more insightful alerts, with fewer false positives and better investigation. In the case of regulators, the system will provide anonymised and aggregated reports of system risk pattern like increased in layering activity among various institution which can aid in supervision and regulatory responses.

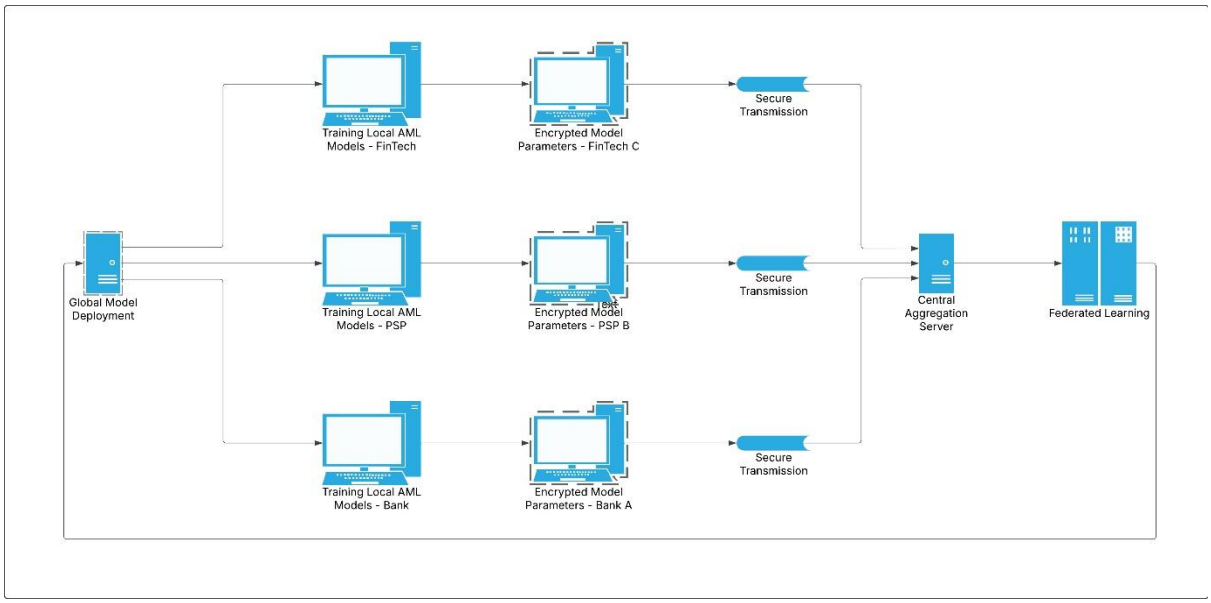


Figure 1

3. Data Access Needs

Functionally, the success of the AML proposed here relies on whether each participating institution has the capabilities to use some type of internal data to train the model. Nevertheless, because of the federated nature of the system, it does not receive or transmit raw data directly. Rather, localised training processes are carried out in secure settings of each financial institutions, and merely encrypted and anonymised updates of the model are shared via secure communication channels. This approach guarantees that the data stays within legal as well as operation boundaries.

For successful model learning, institutions must permit their in-house AML platform to transaction metadata, which includes timestamps, transaction amounts, merchant category codes and basic geographic information. These features are crucial for analysing behavioural patterns and identify potential irregularities. Additionally, institutions can leverage their internal customer level risk indicators like KYC onboarding or past monitoring alerts which can augment supervised learning and improve the model's applicability in real world financial crime patterns. Further, the results from past alerts from investigations, true or false positives and records can serve as labels to informed learning targets and model precision.

These data pools are intended to exist only on the IT infrastructure of participating institutions. The system will use a secure, containerised training environment set up to be executed locally. Activities such as model training, feature engineering and internal data pre-processing will occur within the firewall of the institution so that no raw data is ever exposed or sent at any stage of the process. The only information to be sent outside are the encrypted model parameters that contain no context adequate for the te-identification of individuals or reconstructs of transaction records.

The system will fully adhere to the Data Protection Act of 2018 and UK GDPR. This legal framework acknowledges the significance of data minimisation, purpose limitation and data protection throughout the processing stages. By ensuring that all processing occurs locally and that outputs are anonymised, the system satisfies these regulations while also fulfilling the requirements of Money Laundering Regulations 2017.

Importantly, the system fundamentally operates on premises to maintain sole ownership and authority over their data and will ever be required to reveal it to the central aggregator. This design enables scalability in a heterogeneous fintech environment while allowing for regulators adherence and operational independence.

4. Data Privacy and Governance

The foundation of the system's privacy lies in the federated learning architecture. The distributed approach guarantees that data never leaves the environment of the institution that owns the data and never exchange raw data. Rather, local model training is carried out, the parameters are always encrypted during exchange. To enhance protection against inference attacks, differential privacy mechanisms are introduced prior the data leaving the local system. These methods will add statistical noise to combined outputs to rendering it mathematically unlikely to deduce any individual data point (Dwork, 2006). Moreover, the secure multi-party computation protocols are used while aggregating the models, where the central server is not able to see or manipulated any of the encrypted data. All interactions between local clients are encrypted with safeguarding protocols which had a certificate-based authentication to block unauthorised access.

From the perspective of policy and governance, the system will adhere to a strict data minimisation. It does not retain raw data like customer detail and transaction records. The encrypted model parameters are only temporarily held only for the aggregation model to process and there after delete the data and ready for redistribution. To enhance transparency and traceability, the system optionally features an immutable audit log system which is a blockchain technology to document timing, source and versions of the model updates. This log is completely anonymised and cryptographically signed, allowing institutions to confirm their involvement and audit the lifecycle of their model without disclosing sensitive information.

The platform's architecture aligns with the core requirements established by the UK General Data Protection Regulation (GDPR), including aspects related to data security, integrity, accountability, and protection of individual rights. The Information Commissioner's Office (ICO) emphasizes the importance of incorporating privacy and governance controls into any system handling sensitive personal data, particularly when dealing with data sharing or machine learning (ICO, 2023). The AMLaaS platform complies with this principle by avoiding centralized data processing, thereby reducing its exposure to purely non-sensitive, derived data, while at the same time allowing the institutions holding the data to exert governance control.

Institutions have full control over their participation in the training cycles. They decide what data is used, when to submit model updates, and how to incorporate the returned global model into their internal AML workflows. This decentralized control and accountability not only ensure regulatory compliance but also promote institutional trust and enable interoperability in a fragmented FinTech landscape. Through the use of a combination of rigorous privacy engineering and sound governance design, the AMLaaS platform provides a secure, transparent, and scalable solution for cross-institutional cooperation in AML activities.

5. High Level Architecture

The architecture of the system is structures into three main levels: the nodes for institutional clients, the layer responsible for coordinating federated learning, and the layer for secure aggregation. Each of these layers serves a distinct role in the overall framework of federated model training, aggregation, and sharing.

The first layer is referred to as the client nodes which is implemented on-site at each of the participating institutions. This layer includes the local repository of the institution which are the transaction records, a local feature module, and a containerized federated learning model. The responsibility of the local model includes loading the institution's pre-approved datasets, transforming them suitable for the model, training according to an established AML learning objective, and producing encrypted model updates. This entire procedure is conducted exclusively within the institution's environment, therefore, ensuring that no sensitive customer information is shared or disclosed. The model incorporates compliance modules intended to maintain internal access controls, ensure immutability of logs, and facilitate integration with the institution's risk management systems.

The next layer is referred to as the coordination layer which plays an important role in federated learning by monitoring the interactions between all the participating nodes (institutions) and the central aggregation system. This component utilises a secure and encrypted messaging system to manage the scheduling of training rounds while also concurrently performing validation checks and authenticating updates. Institutions have the flexibility to either participate or skip specific rounds

depending on their availability or timelines. The coordination layer is structured to allow institutions to participate regardless of differences in their operating schedules.

The architecture is fundamentally anchored by a secure aggregation layer, which is designed to run a cloud-based environment which is aimed at privacy enhancement. This layer receives the encrypted model parameters from institutional nodes and performs federated averaging using secure multi-party computation protocols. Importantly, no single party involved and not even the main operator, has the ability to inspect any or all individual contributions. Once the aggregation layer is completed, the new global model is validated before it is spread to all participating institutions. As an optional feature, an audit logging service (blockchain) will be used to record all model updates, timestamps, and participating institution identities in anonymized forms, therefore supporting regulators and internal compliance teams in verifying contributions and maintaining the model's integrity.

Furthermore, the system includes features that are user-friendly interface for the institutions, which facilitates access to training results, schedule modifications, and various other configuration options. This interface supports model versioning and enables each institution to analyse and evaluate the risk signals produced by the federated model and by doing that allows for them to adapt it for integration with their own respective internal transaction monitoring systems.

Figure 2 illustrates the architectural configuration through a straightforward form of a simple block diagram. The configuration encompasses the entire model lifecycle, which includes local training, secure communication, secure aggregation, and final redistribution. Each institutional node runs independently and autonomously, interacting solely via interfaces that are designed to maintain privacy. As shown in the diagram, the platform has no access to sensitive data, thus complying with GDPR principles and fostering regulatory trust.

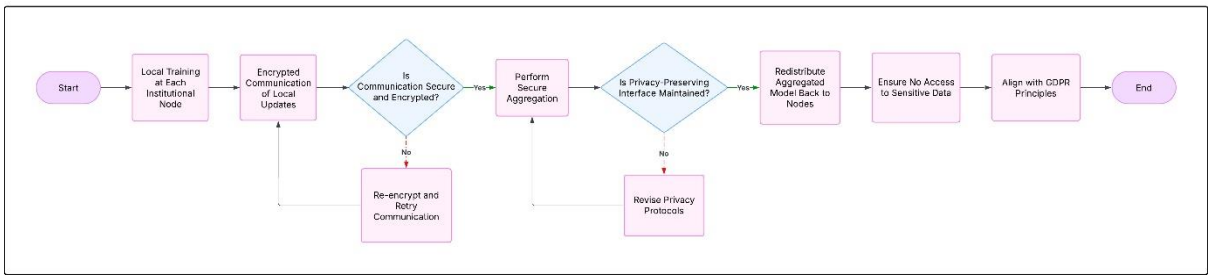


Figure 2

6. Tools and Methodologies

The AMLaaS system serves as a structured framework supported by a technological environment which is aimed at addressing specific challenges encountered during different stages of the ML cycle. The primary focus of the system is on the layering stage, where all the illicit funds are directed through complex networks of transactions to conceal their source. In addition, the system provides additional support during the initial detection at the placement stage and also in subsequent identification of patterns during the integration stage where especially high-volume transaction like banking and electronic payment systems.

At the foundation of the system is the federated machine learning, which facilitates the collaborative model development among various institutions while safeguarding sensitive information. Each participating institutions operates a Python-based system, like PyTorch or TensorFlow Federated, to create classifiers based on either tree or neural network which are trained in transaction data related

to the specific institution. The data preparation process incorporates feature engineering targeted towards AML applications which include metrics like number of transactions, transaction amount, and use of device anomalies. Prior to sending these features to the training models they will be required to undergo techniques like standardisation and normalisation using Pandas, Scikit-learn or Spark in case of large institutions.

Once features are generated, the system advances from the local training stage, where organizations implement supervised learning techniques using labelled historical transaction data. This data encompasses all previous Suspicious Activity Reports (Sars) and will identify all false positives and escalation records. The model architecture may include gradient boosting classifiers like XGBoost or LightGBM, alongside deep learning models like Long Short-Term Memory (LSTM) networks tailored for time-series data, depending on the volume of the data and associated risk classification applicable to the institution. Organizations with access to behavioural attributes like login or device-switching can also include their own temporal models to track changes in behaviour over a period of time which is also an important indicator in the layering activities.

Following the training phase, institutions can implement differential privacy techniques on the model parameters that they have obtained. Some tools like Opacus are very compatible with PyTorch and TensorFlow Privacy which add mathematically specified noise, ensuring that no single transaction can be discerned from the modifications that are made. The differentially private weights are then encrypted through secure multiparty computation protocols and sent to the central aggregation node where it uses federated averaging to aggregate the updates to arrive on a global model capturing the laundering behaviours over the participants and protecting the confidentiality of institutional traits.

In the global context of the model redistribution is essential for achieving uniformity AML detection capabilities of different institutions. Once the latest model is acquired, every institution is capable of using internal scoring pipelines to apply the model to real-time transactions, utilising near real-time processing or batch processing techniques. The alerts produced by the federated model are analysed along with institution-specific risk frameworks to determine whether they need further investigation or are required to be rejected as false positives. In addition, the integration can include explainable artificial intelligence tools, which can help explain the features that had a significant to the impact on the risk alert, therefore helping various analysts to prioritize their reviews more effectively.

Beyond transaction risk assessment, financial institutions can employ federated feature representations for generating scenarios and detecting anomalies. Unsupervised machine learning methods like autoencoders and k-means clustering, can be integrated into the local processing model to examine and investigate new ML tactics. These methods enhance and improves detection during the integration phase of the laundering process, in which the integration of illicit funds into legitimate business operations, a stage where traditional systems often prove to be less effective and often fall short.

Throughout its operational lifespan, the system creates a self-sustaining feedback loop. The participating institutions classify, assess and evaluate identified transactions, then reintegrate verified procedures back into the local structure. The feedback loop can be used to modify the local model's relevance or inform the future iterations of federated training. This enables the platform to improve its ability to respond illicit activities without depending on a centralized training dataset or regulatory authorities.

7. Conclusion

The Anti Money laundering As A Service system utilised advanced analytical methodology, privacy-preserving strategies, and adaptive learning techniques to enhance detection throughout various phases of the ML process. By focusing on the data collections and pre-processing, collaborative training and implementation, this solution showcases a structure, systematic and scalable approach to tackling the increasing challenges of AML compliance within the digital economy of the United Kingdom.

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